Program: B Tech & MBA Tech (Computer Engineering, Information Technology, Artificial Intelligence) B Tech (Computer Science, AIML) BTI Computer Engineering	**Semester:** VI /VII /V / VIII/X
Course: Distributed Computing	**Code:**702CO0C034

Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks - 100)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Prerequisite: Operating Systems

Course Objective

To introduce the concepts and design of distributed computing and algorithms that support distributed computing.

Course Outcomes

After completion of the course, student will be able to -

1. Explain the basic concepts of distributed computing
2. Apply the concepts of distributed computing to implement various mechanisms of communication
3. Analyze various approaches of synchronization, mutual exclusion, election algorithms and fault tolerant services
4. Recognize different kinds of naming and their implementation

Detailed Syllabus

Unit	Description	Duration
1	**Introduction** to **Distributed System** Definition, Goals, Examples of Distributed System-Internet. System architectures-centralized architecture, decentralized architecture, hybrid architecture, Client-Server Model, Servers-general design issues, server clusters, managing server clusters.	**05**
2	**Communication** Basic RPC operation, RPC implementation, RPC semantics in presence of failures, RMI- Basics, Implementation, Case study-	**06**

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	Java RMI, Message oriented communication-: transient and persistent communication. Stream oriented communication- support for continuous media, streams and QoS, stream synchronization.	
3	**Synchronization** Introduction, Physical Clock synchronization algorithms, Logical clocks, event ordering, implementation of Logical clocks, Lamport's logical clock algorithm, Vector clocks, Mutual exclusion: Centralized, distributed and token ring mutual exclusion algorithms, comparison of these algorithms. Traditional election algorithm- Bully and Ring election algorithm.	**06**
4	**Fault Tolerance** Introduction, Process resilience, Reliable group communication.	**08**
5	**Naming** Names, identifiers, and addresses, Flat naming , Structured naming: name spaces and resolution, implementation of name space, Case study- Domain Name System, Attributed based naming- Directory services.	**05**
	Total	**30**

Text Books

1. Andrew S. Tanenbaum, *Distributed System: Principles and Paradigms*, 3rd Edition, Pearson Prentice Hall, 2017. ISBN 9789332549807

Reference Books

1. George Couloris, *Distributed System: Concept and Design*, 5th edition, Pearson Education, 2017. ISBN 9789332575226
2. Pradeep K. Sinha, *Distributed Operating System*, IEEE Press, Prentice Hall of India Ltd, 1998.
3. Mei-Ling L. Liu, *Distributed Computing: Principles and Applications*, Addison – Wesley, Publisher : Pearson Education, 2004 ISBN-13 : 978-8131713327.

Laboratory / Tutorial work:

8 to 10 experiments (and a practicum where applicable) based on the syllabus.

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